



Session 9C: 3D Geometry — Surfaces, Distance, and Space

Phase 2 — Classical Techniques | 75 min

Extending 2D intuition into three dimensions. No calculus required.

Part A: Points, Planes, and Distance in 3D

Example 1: Distance Between Two Points in 3D

In 2D: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

In 3D, add the z -dimension:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

$$(1, 2, 3) \text{ to } (4, 6, 15): d = \sqrt{3^2 + 4^2 + 12^2} = \sqrt{9 + 16 + 144} = 13.$$

This generalizes to any dimension: in $\mathbb{R}^n$, $d = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$.

Example 2: The Equation of a Plane

A plane in 3D is the analog of a line in 2D. A line is $ax + by = c$. A plane is $ax + by + cz = d$.

$2x + 3y - z = 6$: at $x = 0, y = 0 \rightarrow z = -6$. At $y = 0, z = 0 \rightarrow x = 3$. At $x = 0, z = 0 \rightarrow y = 2$.

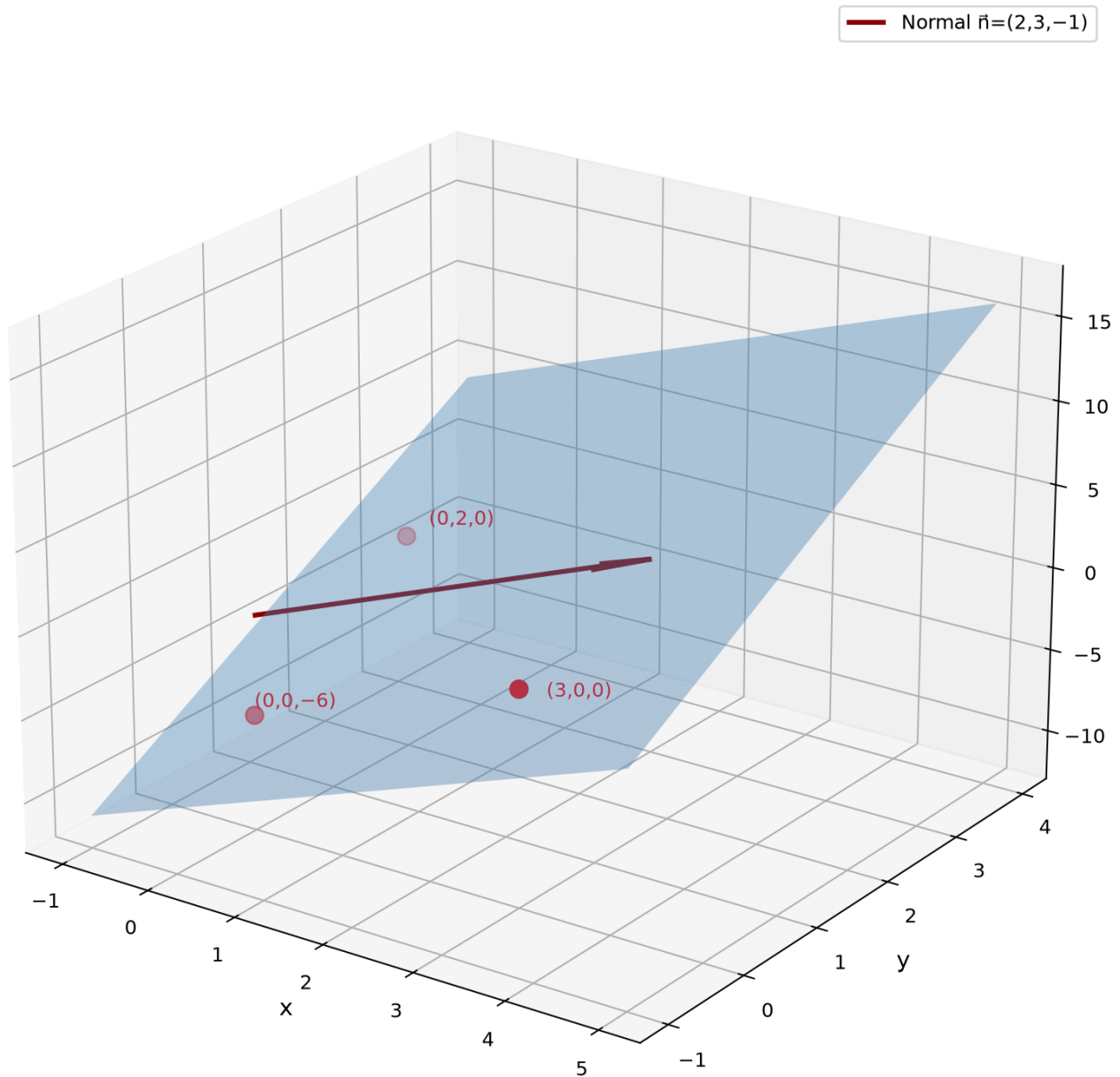
The plane passes through $(3, 0, 0)$, $(0, 2, 0)$, $(0, 0, -6)$.

Normal vector: $\vec{n} = (a, b, c) = (2, 3, -1)$. The normal is perpendicular to every direction lying in the plane.

Plane from point and normal: $\vec{n} \cdot (\vec{x} - \vec{p}) = 0$.

Through $(1, 2, 3)$ with normal $(2, -1, 1)$: $2(x - 1) - (y - 2) + (z - 3) = 0 \rightarrow 2x - y + z = 3$.

**Graph 9C: A Plane in 3D — $2x + 3y - z = 6$
Intercepts + Normal Vector**



Graph 9C: The plane $2x+3y-z=6$ showing its three intercepts $(3,0,0)$, $(0,2,0)$, $(0,0,-6)$ and the normal vector $\vec{n}=(2,3,-1)$ perpendicular to the plane.

Example 3: Distance from a Point to a Plane

Point (x_0, y_0, z_0) to plane $ax + by + cz = d$:

$$D = \frac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2 + b^2 + c^2}}.$$

$(1, 2, 3)$ to $2x + 3y + z = 6$:

$$D = \frac{|2+6+3-6|}{\sqrt{4+9+1}} = \frac{5}{\sqrt{14}} \approx 1.336.$$

Geometric meaning: The numerator measures how far the point "misses" the plane equation. The denominator normalizes by the length of the normal vector.

Example 4: The Sphere — All Points at a Fixed Distance

$(x - h)^2 + (y - k)^2 + (z - \ell)^2 = R^2$. Center (h, k, ℓ) , radius R .

$x^2 + y^2 + z^2 = 9$: center $(0, 0, 0)$, radius 3. All points exactly 3 units from the origin.

Complete the square for $x^2 + y^2 + z^2 - 4x + 6y - 2z - 11 = 0$:

$(x - 2)^2 + (y + 3)^2 + (z - 1)^2 = 11 + 4 + 9 + 1 = 25$. Center $(2, -3, 1)$, radius 5.

Example 5: Distance from a Point to a Sphere

Point P to sphere center C , radius R :

Distance = $||PC| - R|$.

$(10, 0, 0)$ to sphere $x^2 + y^2 + z^2 = 25$: $|PC| = 10$, $R = 5$. Distance = 5.

The closest point on the sphere is $(5, 0, 0)$ — along the line from center to P .

If the point is inside the sphere, distance = $R - |PC|$.

Part B: Functions of Two Variables — $z = f(x, y)$

Example 6: A Surface as a Height Map

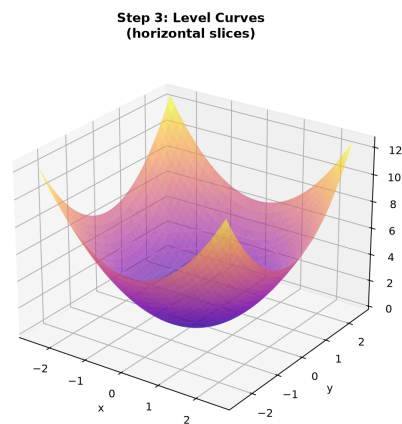
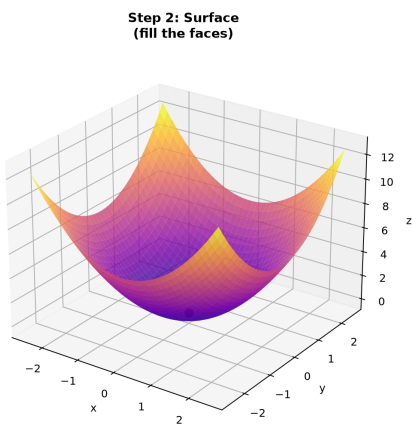
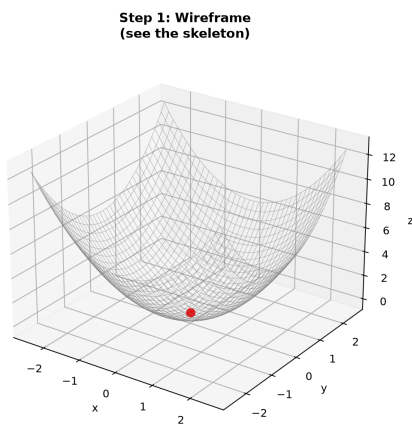
$z = f(x, y)$ assigns a height to every point (x, y) in the plane.

$z = x^2 + y^2$: at $(0, 0) \rightarrow z = 0$. At $(1, 0) \rightarrow z = 1$. At $(2, 1) \rightarrow z = 5$.

The surface is a bowl opening upward — a **paraboloid**.

$z = \sqrt{1 - x^2 - y^2}$: only defined when $x^2 + y^2 \leq 1$. The surface is the **upper hemisphere** of radius 1.

Graph 9C: Building a 3D Surface — $z = x^2 + y^2$



Graph 9C-S1: Building $z = x^2 + y^2$ in three stages. Step 1 — Wireframe skeleton reveals the bowl shape's underlying grid. Step 2 — Fill in the faces to see the solid paraboloid. Step 3 — Add white level curves (horizontal slices at $z=1,2,3,4$) to see how the bowl expands. The red/black dot marks the minimum at $(0,0,0)$.

Example 7: Domain in 3D — Regions in the xy -Plane

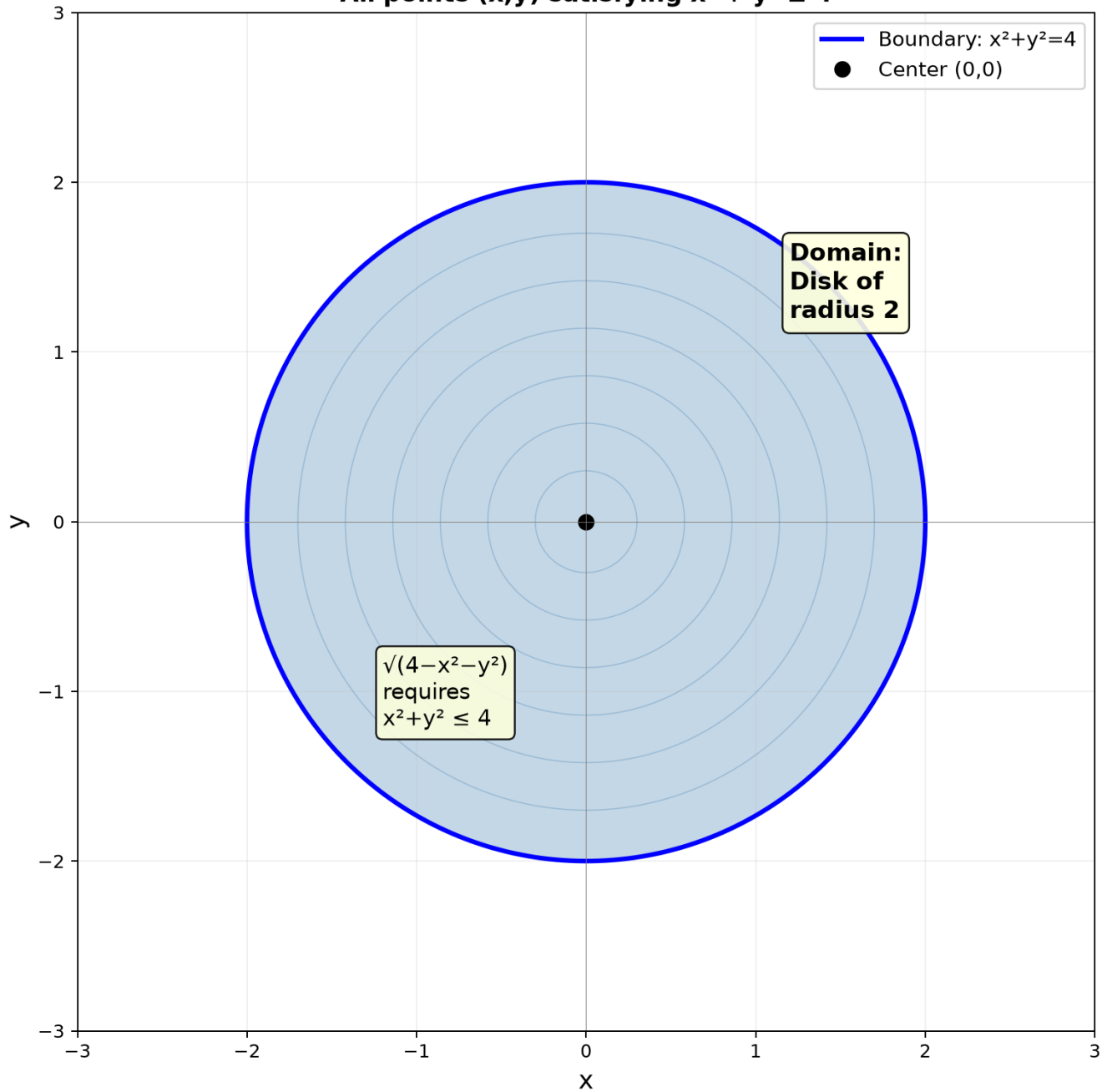
For $z = f(x, y)$, the domain is a region in the xy -plane.

$z = \sqrt{4 - x^2 - y^2}$: needs $x^2 + y^2 \leq 4 \rightarrow$ disk of radius 2.

$z = \ln(x + y)$: needs $x + y > 0 \rightarrow$ half-plane above the line $y = -x$.

$z = \frac{1}{x^2 + y^2 - 1}$: needs $x^2 + y^2 \neq 1 \rightarrow$ entire plane except the unit circle.

Graph 9C6: Domain of $z = \sqrt{4 - x^2 - y^2}$
All points (x,y) satisfying $x^2 + y^2 \leq 4$



Graph 9C6: The domain of a 3D surface is a 2D region. Here the disk $x^2 + y^2 \leq 4$ — the set of all (x,y) where $z = \sqrt{4 - x^2 - y^2}$ is real.

Example 8: Level Curves — A 2D Map of a 3D Surface

A **level curve** is the set of (x, y) where $f(x, y)$ equals a constant c :

$f(x, y) = c$. Think of it as a horizontal slice through the surface at height c .

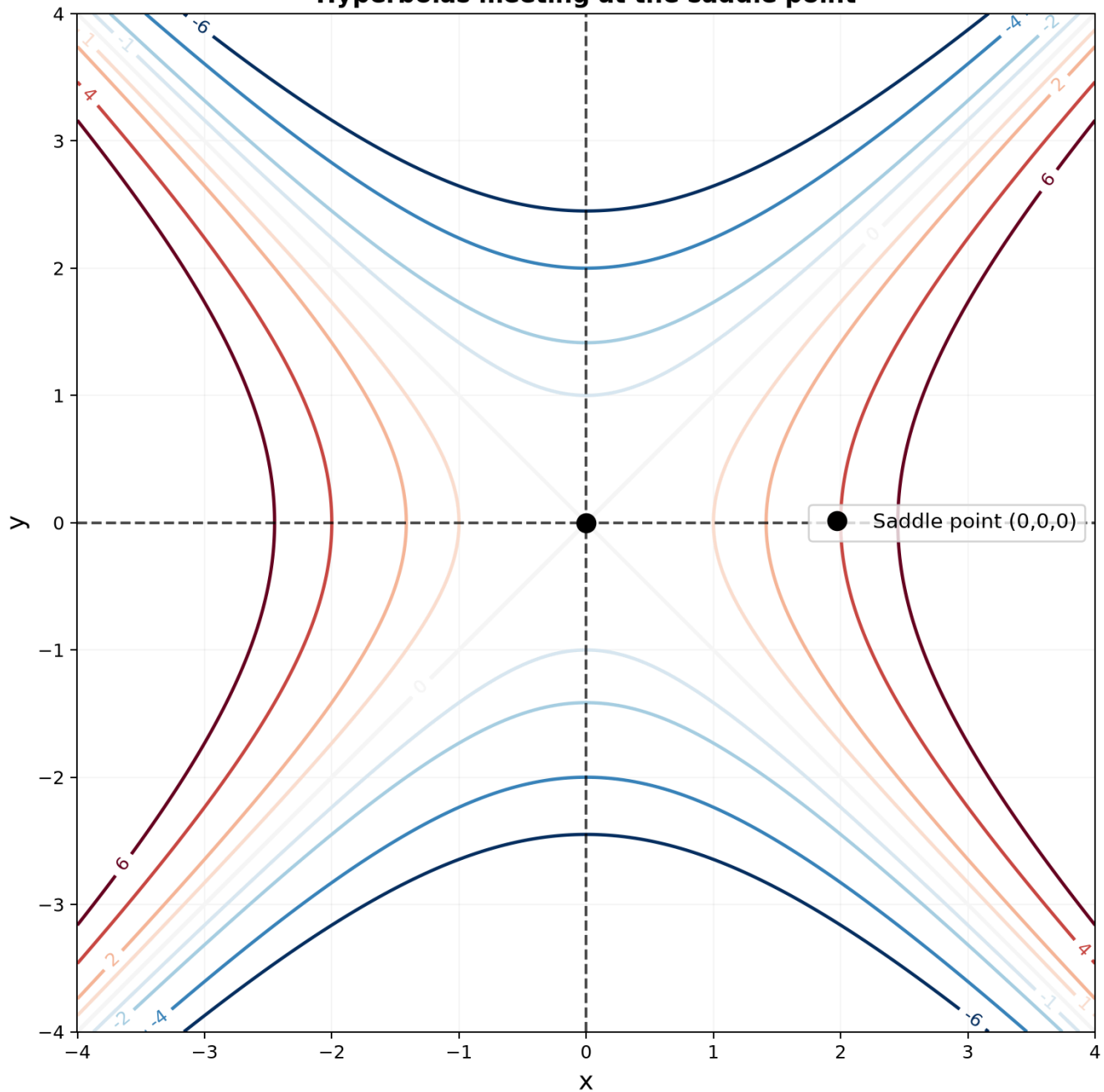
$z = x^2 + y^2$: level curves are circles $x^2 + y^2 = c$. As c increases, circles grow — the bowl expands.

$z = y - x^2$: level curves are parabolas $y = x^2 + c$. Shifting up/down.

$z = xy$: level curves are hyperbolas $xy = c$. For $c > 0$, Q1/Q3. For $c < 0$, Q2/Q4.

How to read: If level curves are close together, the surface is steep. If far apart, the surface is flat.

Graph 9C1: Level Curves of $z = x^2 - y^2$
Hyperbolas meeting at the saddle point



Graph 9C1: Level curves of $z=x^2-y^2$ — hyperbolas for $z \neq 0$, crossing lines for $z=0$. The red/blue colors indicate positive/negative heights. The saddle point at $(0,0,0)$ is where the crossing lines meet.

Part C: Quadric Surfaces — The Six Classic 3D Shapes

In 2D, conic sections give circle, ellipse, parabola, hyperbola. In 3D, quadric surfaces are the analogous family.

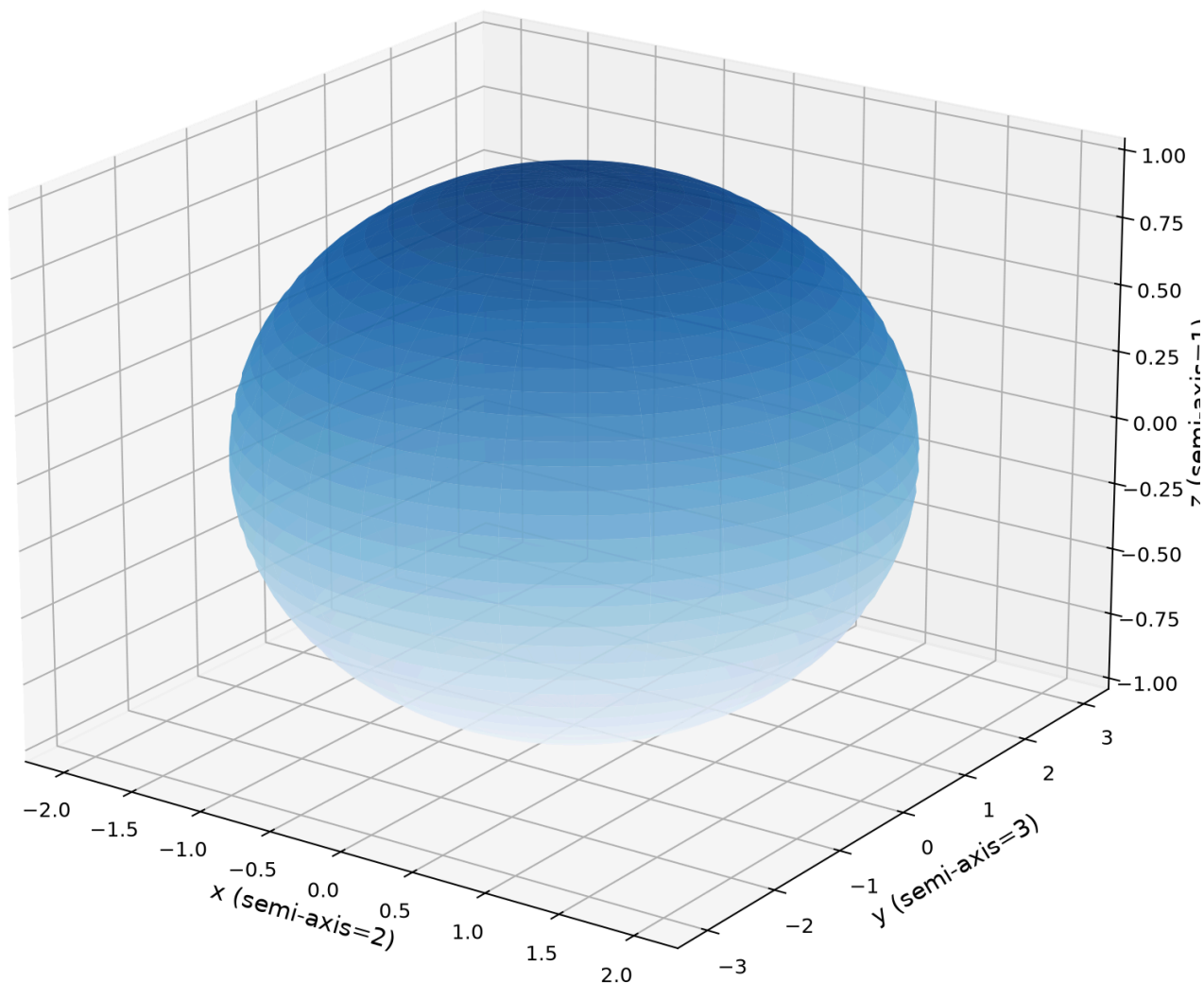
Example 9: Ellipsoid — The 3D Ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

All three cross-sections are ellipses. If $a = b = c$, it's a sphere. If $a = b \neq c$, it's a squashed sphere (spheroid).

$$\frac{x^2}{4} + \frac{y^2}{9} + \frac{z^2}{1} = 1: \text{ extends } \pm 2 \text{ in } x, \pm 3 \text{ in } y, \pm 1 \text{ in } z.$$

Graph 9C2: Ellipsoid — $x^2/4 + y^2/9 + z^2/1 = 1$



Graph 9C2: An ellipsoid with semi-axes $a=2$, $b=3$, $c=1$. All three cross-sections perpendicular to the axes are ellipses. When $a=b=c$, it becomes a sphere.

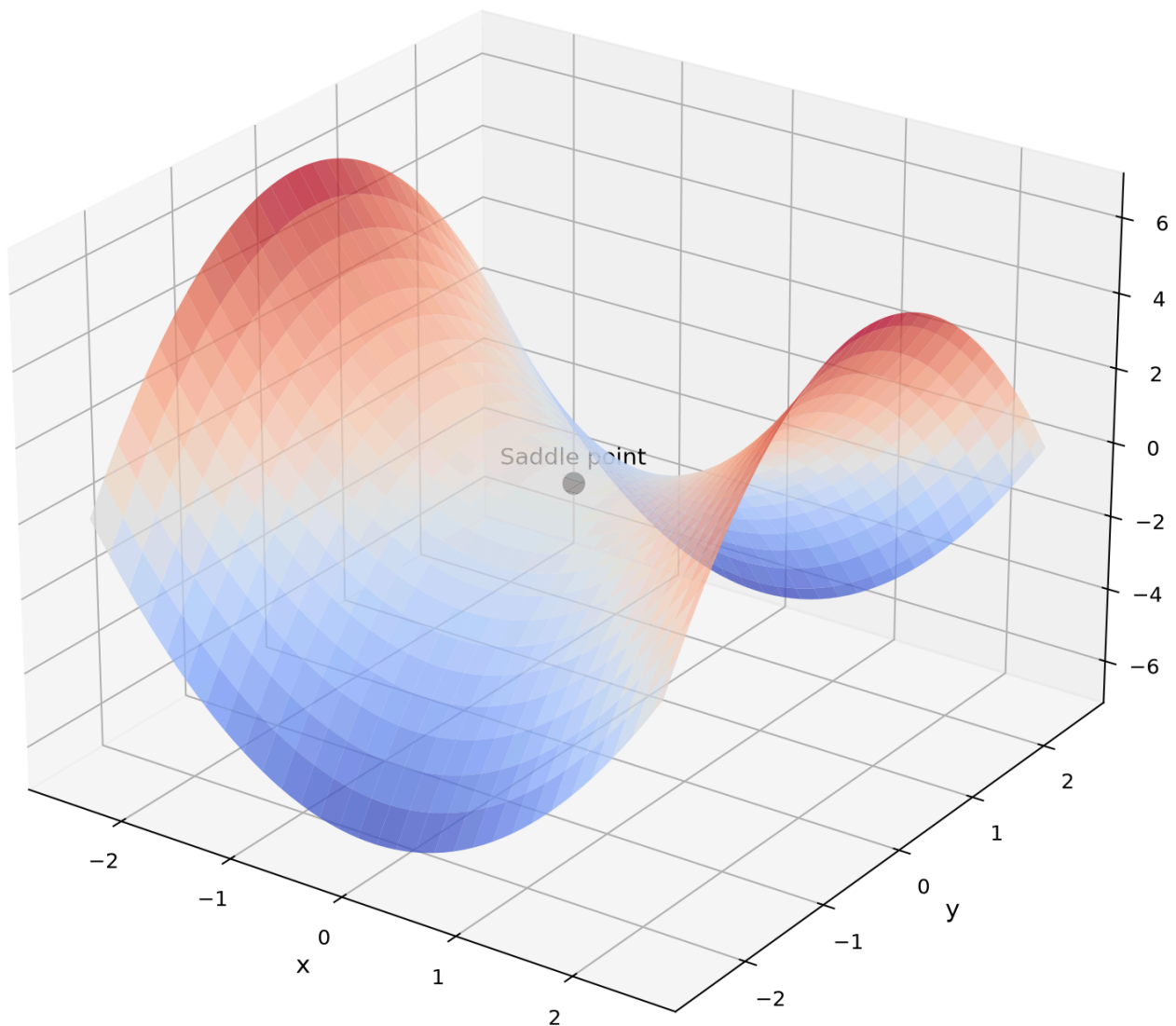
Example 10: Paraboloid — The 3D Parabola

Elliptic paraboloid: $z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$. Bowl shape. All vertical cross-sections are parabolas.

$z = x^2 + y^2$: circular paraboloid. $z = x^2 + 2y^2$: elliptical paraboloid (stretched in y).

Hyperbolic paraboloid (saddle): $z = \frac{x^2}{a^2} - \frac{y^2}{b^2}$. Curves up in one direction, down in the other.
Shaped like a Pringle chip.

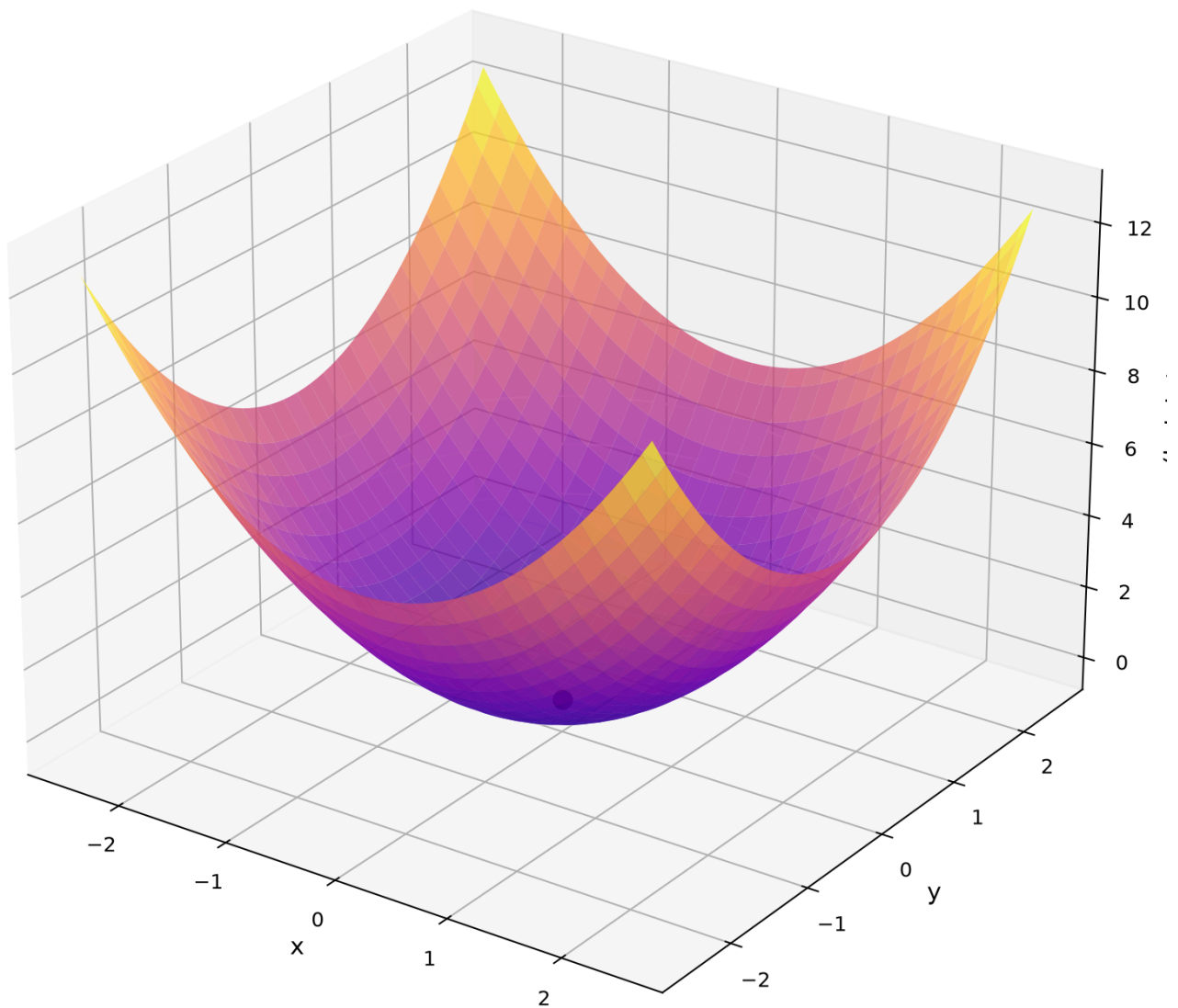
Graph 9C: Hyperbolic Paraboloid (Saddle)
 $z = x^2 - y^2$ — curves up in x , down in y



Graph 9C: The hyperbolic paraboloid $z=x^2-y^2$ — a saddle. It curves upward along the x -direction (a parabola opening up) and downward along the y -direction (a parabola opening down). The

origin is the saddle point.

Graph 9C5: Elliptic Paraboloid $z = x^2 + y^2$
Level curves are circles $x^2 + y^2 = c$ (white rings)



Graph 9C5: The bowl-shaped elliptic paraboloid $z = x^2 + y^2$. White rings show level curves (circles). The vertex is at the origin. Every vertical cross-section is a parabola.

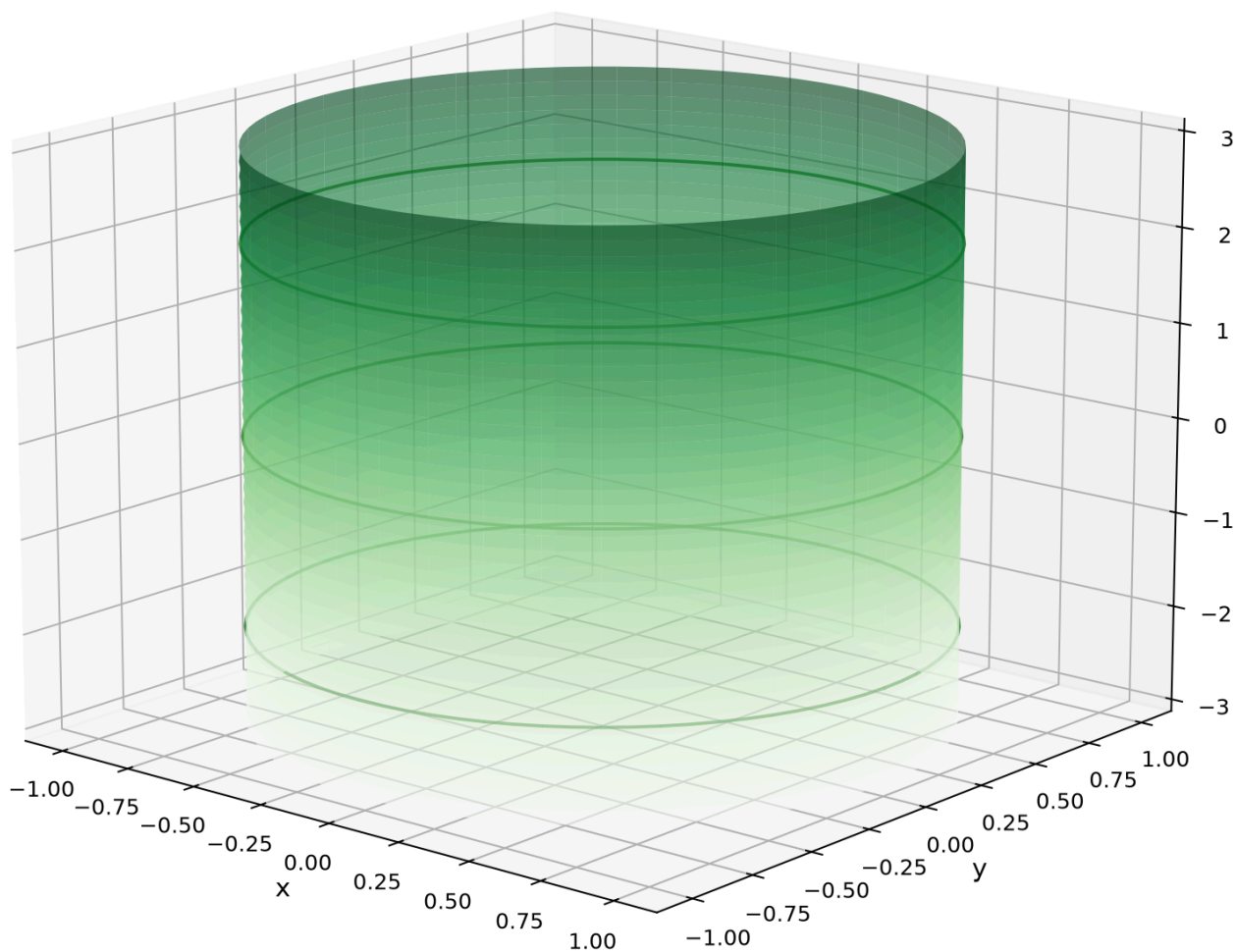
Example 11: Cylinder — Extruding a 2D Curve

A cylinder has one variable missing from the equation.

$x^2 + y^2 = 9$: in 3D, this is a circular cylinder — a circle of radius 3 extruded infinitely along the z -axis. z can be anything.

$z = \sin x$: in 3D, a wavy sheet extruded along the y -axis. Every vertical slice parallel to the xz -plane shows the same sine wave.

Graph 9C: Circular Cylinder $x^2 + y^2 = 1$
A circle extruded infinitely along z



Graph 9C: The cylinder $x^2 + y^2 = 1$ — a circle of radius 1 extruded infinitely along the z-axis. The horizontal rings show that every cross-section perpendicular to z is the same circle.

Example 12: Hyperboloid — One Sheet or Two

One sheet: $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$. Shaped like a cooling tower. Connected.

Two sheets: $-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$. Two separate bowls opening away from each other along the z -axis.

Identifying: One sheet = two plus signs, one minus. Two sheets = two minus signs, one plus.

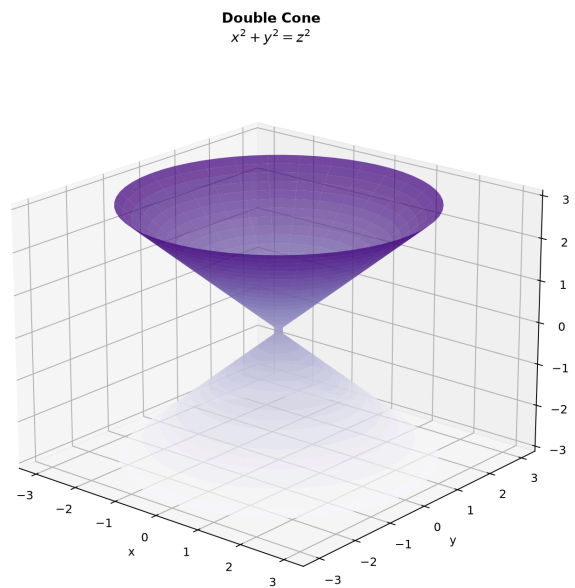
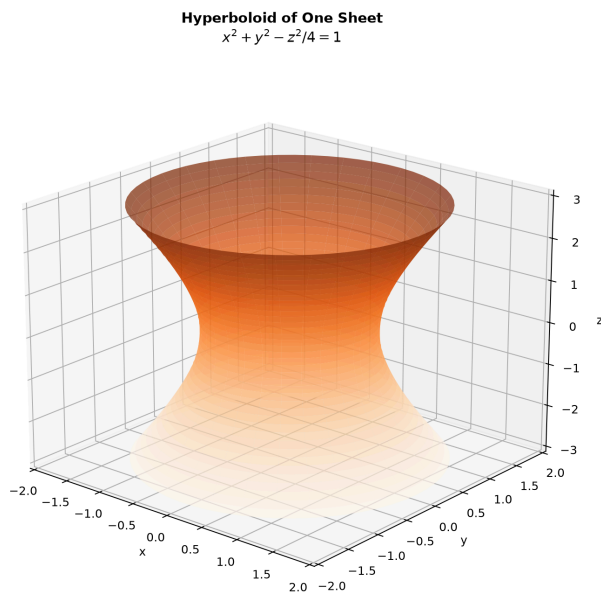
Example 13: Cone — The Double Nappe

$z^2 = x^2 + y^2$. Two cones meeting at the origin: one opening upward, one downward.

At $z = k$, the cross-section is a circle $x^2 + y^2 = k^2$. The radius grows linearly with height.

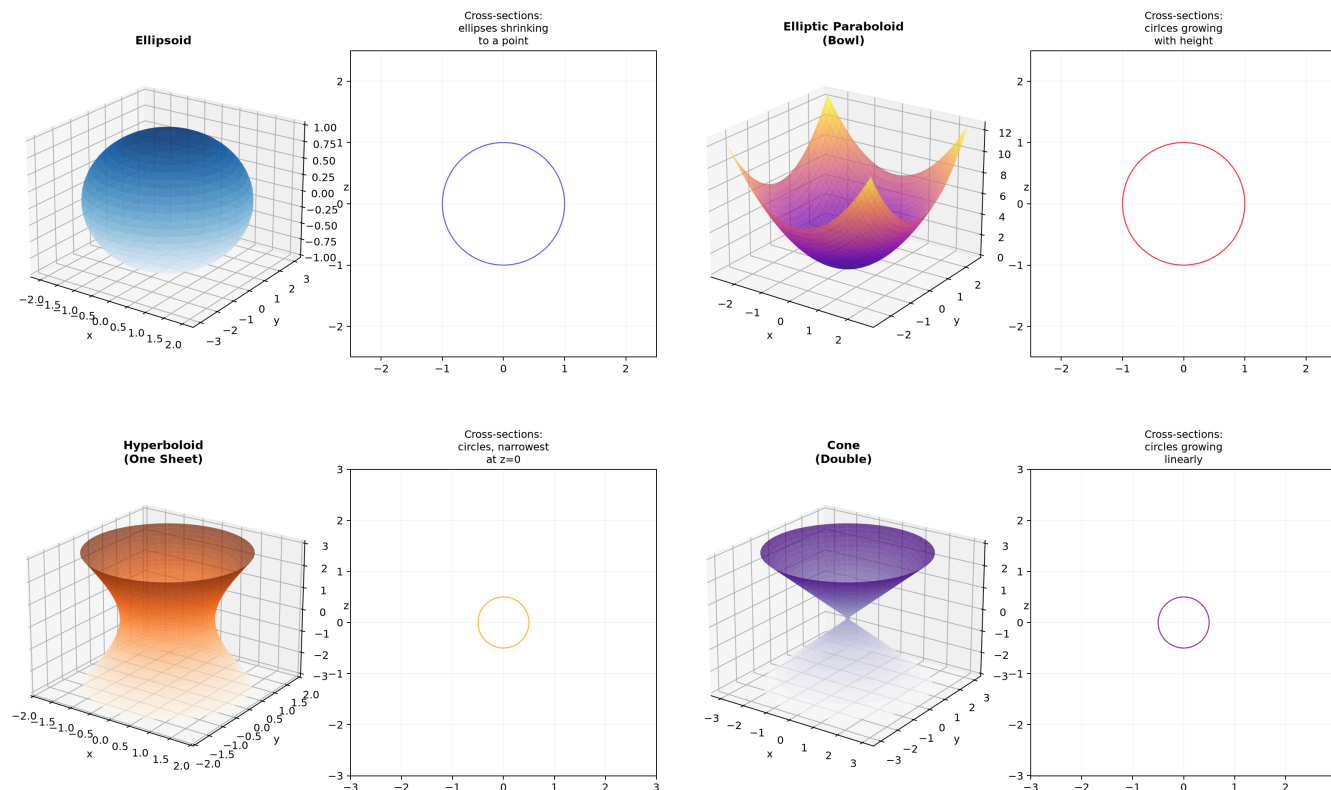
Elliptic cone: $\frac{z^2}{c^2} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$. The cross-section is an ellipse instead of a circle.

Graph 9C3: Hyperboloid of One Sheet vs. Double Cone



Graph 9C3: Left — Hyperboloid of one sheet ($x^2 + y^2 - z^2/4 = 1$), shaped like a cooling tower. Right — Double cone ($x^2 + y^2 = z^2$), two nappes meeting at the origin.

Graph 9C: Quadric Surface Gallery — 3D Shapes + Cross-Sections



Graph 9C-S2: The four main quadric surfaces side by side — Ellipsoid, Elliptic Paraboloid (bowl), Hyperboloid of one sheet (cooling tower), and Double Cone. Right column — What you see when you slice each horizontally: ellipses that shrink, circles that grow, circles narrowest at $z=0$, and circles growing linearly. Understanding cross-sections makes these 3D shapes predictable.

Part D: Intersection and Symmetry in 3D

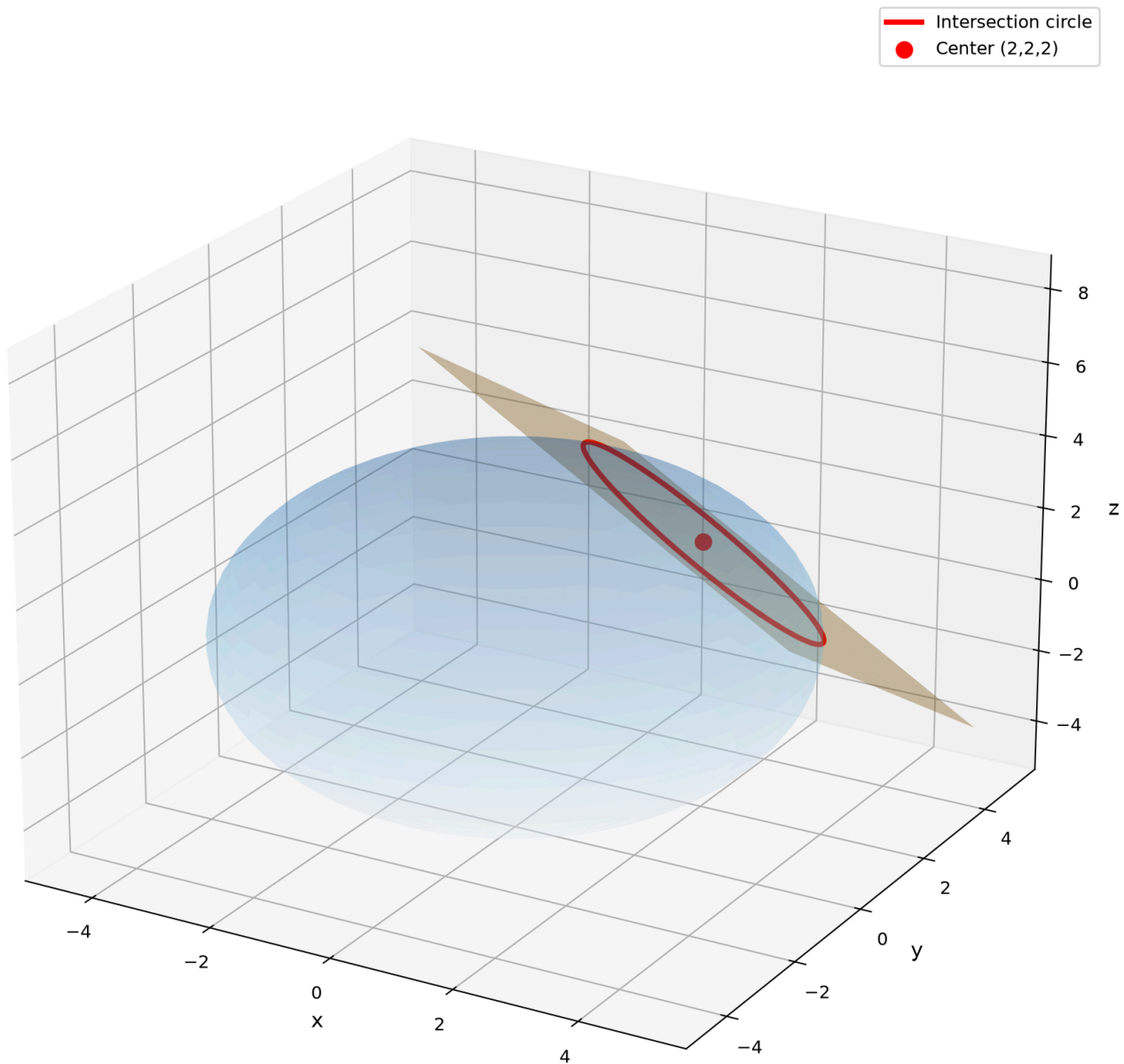
Example 14: Intersection of Two Surfaces — A Curve in Space

Intersect the sphere $x^2 + y^2 + z^2 = 25$ with the plane $x + y + z = 3$:
The result is a circle in the plane.

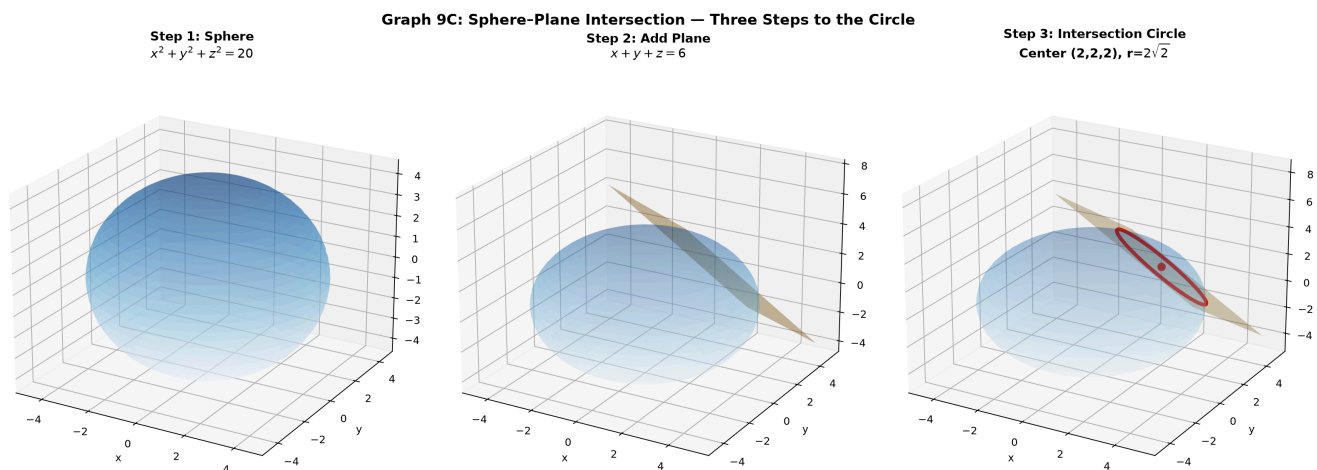
How to find center and radius of the intersection circle:

- (1) Distance from origin to plane: $d = \frac{|3|}{\sqrt{3}} = \sqrt{3}$.
- (2) Radius of intersection: $r = \sqrt{R^2 - d^2} = \sqrt{25 - 3} = \sqrt{22}$.
- (3) Center is the foot of the perpendicular: $(1, 1, 1)$.

**Graph 9C4: Sphere $x^2+y^2+z^2=20$ $\cap$ Plane $x+y+z=6$
Intersection: circle, center $(2,2,2)$, radius $2\sqrt{2}$**



Graph 9C4: The sphere $x^2+y^2+z^2=20$ and plane $x+y+z=6$ intersect in a circle (red). The center $(2,2,2)$ lies along the normal direction from the origin to the plane.



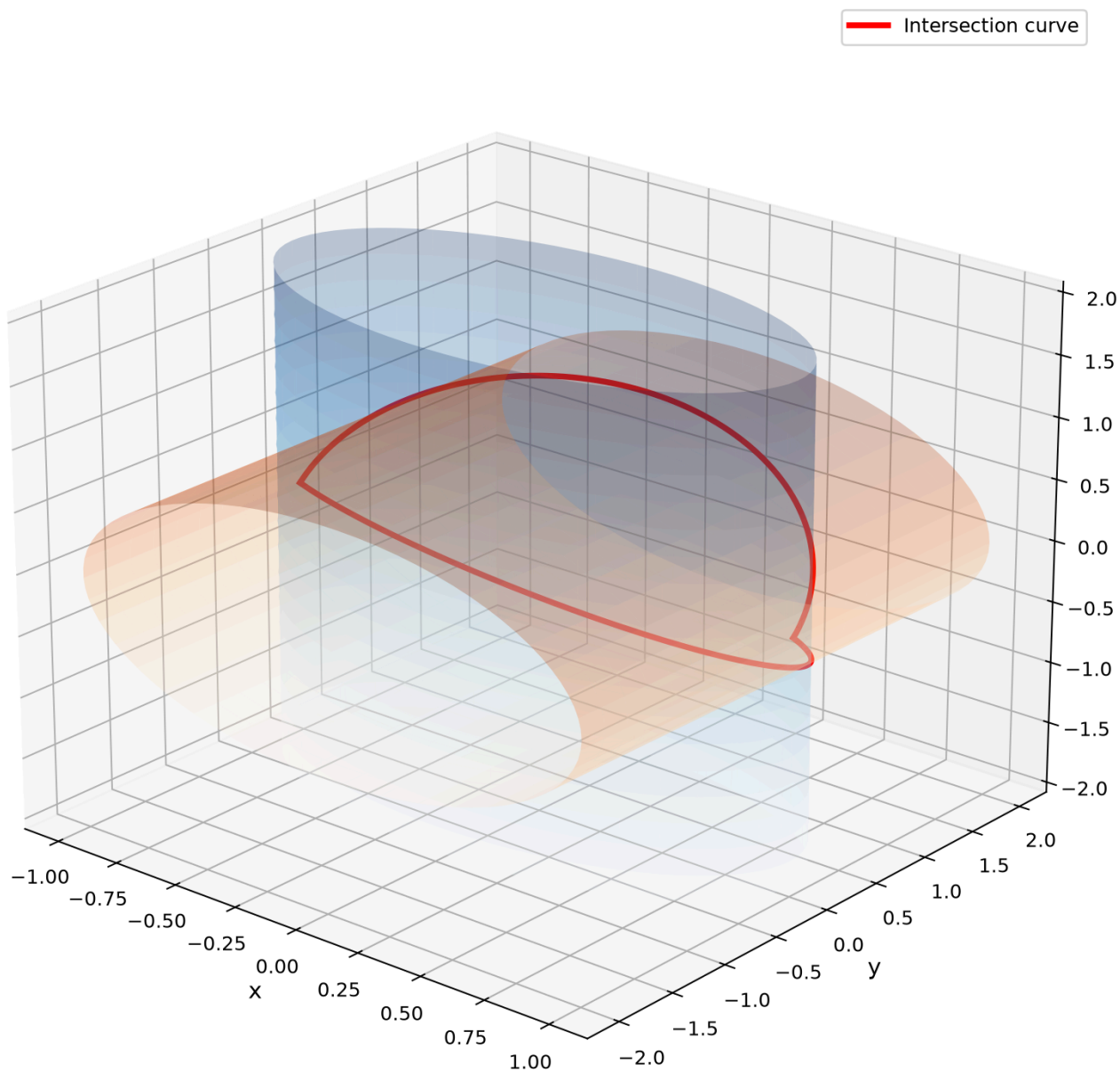
Graph 9C-S3: Finding the intersection circle in three steps. Step 1 — The sphere $x^2 + y^2 + z^2 = 20$ alone. Step 2 — The plane $x + y + z = 6$ slices through it (orange). Step 3 — The intersection emerges as a red circle, center (2,2,2), radius $2\sqrt{2}$. The key formula: $r = \sqrt{R^2 - d^2}$ where d is the distance from sphere center to plane.

Example 15: Intersection of Two Cylinders

$x^2 + y^2 = 1$ and $x^2 + z^2 = 1$. Two cylinders intersecting at right angles. The intersection curve is a "bicylindrical curve" — it looks like an ellipse when viewed from certain angles.

Graph 9C: Intersecting Cylinders

$$x^2 + y^2 = 1 \text{ and } x^2 + z^2 = 1$$



Graph 9C: Two perpendicular cylinders (blue: $x^2 + y^2 = 1$, orange: $x^2 + z^2 = 1$) intersect in a 3D curve (red). This is a preview of how 3D surfaces can create space curves.

Example 16: Symmetry in 3D

A surface $F(x, y, z) = 0$ has:

- **xy -plane symmetry**: if replacing $z \rightarrow -z$ leaves the equation unchanged.
- **xz -plane symmetry**: if $y \rightarrow -y$ unchanged.
- **Origin symmetry**: if $(x, y, z) \rightarrow (-x, -y, -z)$ unchanged.
- **Rotational symmetry about z -axis**: if $x^2 + y^2$ appears as a block.

$x^2 + y^2 + z^2 = 1$: all three coordinate plane symmetries + origin symmetry + rotational symmetry about any axis through origin (a sphere has maximum symmetry).

$z = x^2 + y^2$: rotational symmetry about z -axis, xz and yz plane symmetries.

Up to here: 3D distance and planes. $z = f(x, y)$ as height maps. Level curves = 2D slices.
Quadric surfaces: ellipsoid, paraboloid, cylinder, hyperboloid (1/2 sheets), cone.
Intersection of surfaces = space curves. Symmetry in 3D.

Common Mistakes

Mistake 1: Treating $x^2 + y^2 = 1$ as a circle only

In 2D: it's a circle. **In 3D:** it's a cylinder (infinitely tall). Always check the dimension you're working in.

Mistake 2: Forgetting the square root in the denominator

Wrong: Distance to plane = $\frac{|ax_0 + by_0 + cz_0 - d|}{a^2 + b^2 + c^2}$.

Right: Divide by $\sqrt{a^2 + b^2 + c^2}$, the length of the normal vector.

Mistake 3: Confusing one-sheet and two-sheet hyperboloids

One sheet: exactly one minus sign in the standard form. **Two sheets:** exactly two minus signs.

The one-sheet hyperboloid is connected; the two-sheet one is disconnected.

What We Just Did

- (1) 3D distance: add the z^2 term. Plane: $ax+by+cz=d$, normal vector (a,b,c) .
Point-to-plane distance formula.
- (2) $z = f(x,y)$: a height map over the xy -plane. Domain is a 2D region.
Level curves: horizontal slices – the 2D map of a 3D landscape.
- (3) Quadric surfaces: ellipsoid (3D ellipse), paraboloid (bowl/saddle), cylinder (extruded curve), hyperboloid (1 or 2 sheets), cone (double).
Each has a signature equation pattern.
- (4) Intersection of surfaces gives a curve in 3D space.
Symmetry: test by sign changes on each coordinate.

Practice 1

Find the distance from $(2, -1, 4)$ to the plane $x + 2y + 2z = 6$.

→ Reference: **Example 3**

Solutions: [Solutions](#)

Practice 2

Find the center and radius of the sphere $x^2 + y^2 + z^2 - 4x + 2y - 6z + 5 = 0$.

→ Reference: **Example 4**

Solutions: [Solutions](#)

Practice 3

Describe and sketch the level curves of $z = x^2 - y^2$ at $c = -2, -1, 0, 1, 2$. What quadric surface is this?

→ Reference: **Example 8, 10**

Solutions: [Solutions](#)

Practice 4

Classify each quadric surface:

(a) $x^2 + y^2 - z^2 = 1$

(b) $z = 4 - x^2 - y^2$

(c) $x^2 + 2y^2 + 3z^2 = 12$

→ Reference: **Examples 9-13**

Solutions: [Solutions](#)

Practice 5

Find the center and radius of the circle formed by intersecting the sphere $x^2 + y^2 + z^2 = 20$ with the plane $x + y + z = 6$.

→ Reference: **Example 14**

Solutions: [Solutions](#)

Practice 6: Real Battle

A plane passes through the three points $(1, 0, 0)$, $(0, 2, 0)$, $(0, 0, 3)$. Find its equation and the distance from the origin to this plane.

→ Reference: **Example 2, 3**

Solutions: [Solutions](#)

Basic Algebra Drill — 3D Geometry (10 Problems)

Pure computation.

- D1.** Find the distance between $(1, -2, 3)$ and $(5, 2, 15)$.
- D2.** Write the equation of a plane through $(2, 1, -3)$ with normal vector $(1, 4, -2)$.
- D3.** Find the center and radius of $x^2 + y^2 + z^2 + 8x - 2y + 10z + 8 = 0$.
- D4.** Identify the surface: $x^2 + y^2 = 16$ (in 3D).
- D5.** Identify the surface: $z = 5 - x^2 - y^2$.
- D6.** Identify the surface: $\frac{x^2}{4} + \frac{y^2}{9} + \frac{z^2}{16} = 1$.
- D7.** Identify the surface: $x^2 - y^2 + z^2 = 0$.
- D8.** Find the distance from $(3, 0, 0)$ to the sphere $x^2 + y^2 + z^2 = 4$.
- D9.** Write the equation of the level curve of $z = x^2 + 4y^2$ at height $z = 4$.
- D10.** Does $z = \sqrt{9 - x^2 - y^2}$ have xy -plane symmetry? xz -plane? Justify.

Solutions: [Solutions](#)

Advanced Algebra Drill — 3D Geometry (10 Problems)

Multi-step geometric reasoning.

- A1.** Find the distance between the parallel planes $2x - y + 2z = 5$ and $2x - y + 2z = -7$.
- A2.** Find the equation of the sphere that has $(2, -1, 3)$ and $(6, 5, -1)$ as endpoints of a diameter.
- A3.** The surface $z = xy$ is a hyperbolic paraboloid. Describe its level curves at $z = -2, -1, 0, 1, 2$ and explain the saddle shape.
- A4.** Find the intersection of the line through $(1, 2, 3)$ with direction $(1, -1, 1)$ and the plane $2x + y - z = 4$.
- A5.** A plane passes through $(1, 1, 1)$ and is perpendicular to both $x + y + z = 1$ and $x - y + 2z = 0$. Find its equation. (Hint: normal is cross product of the two normals.)
- A6.** Find the equation of the cylinder whose base is the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$ and whose rulings are parallel to the z -axis.
- A7.** Two spheres: $x^2 + y^2 + z^2 = 25$ and $(x - 8)^2 + y^2 + z^2 = 9$. Do they intersect? If so, find the equation of the plane containing their circle of intersection.
- A8.** Find the point on the plane $x + 2y + 3z = 13$ closest to the origin. (Hint: the closest point lies along the normal direction.)
- A9.** The surface $x^2 + 4y^2 - z^2 = 0$ is an elliptic cone. Describe the cross-section at $z = 2$ and at $y = 1$.
- A10.** Find the volume of the tetrahedron formed by the coordinate planes and the plane $\frac{x}{2} + \frac{y}{3} + \frac{z}{4} = 1$.

Solutions: [Solutions](#)

Today's Procedure

Step 1: 3D distance = one more squared term. Plane = $ax+by+cz=d$.

The vector (a,b,c) is perpendicular to the plane (the normal).

Point-to-plane distance: $|ax_0+by_0+cz_0-d| / \sqrt{a^2+b^2+c^2}$.

Step 2: $z = f(x,y)$ describes a landscape. Domain is a 2D region.

Level curves ($f(x,y)=c$) show the shape in 2D – close curves = steep.

Step 3: Quadric surfaces = 3D analogs of conic sections. Learn the six:

ellipsoid, paraboloid (elliptic/hyperbolic), cylinder,

hyperboloid (1 sheet/2 sheets), cone.

Each has a signature pattern of x^2 , y^2 , z^2 signs.

Terminology

What we call it	Math term	Notation
plane	plane	$ax + by + cz = d$
normal vector	normal vector	$\vec{n} = (a, b, c)$
sphere	sphere	$(x - h)^2 + (y - k)^2 + (z - \ell)^2 = R^2$
height map / landscape	function of two variables	$z = f(x, y)$
level curve	level curve / contour	$f(x, y) = c$
ellipsoid	ellipsoid	$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$
paraboloid (bowl)	elliptic paraboloid	$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$
saddle	hyperbolic paraboloid	$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}$
cylinder	cylinder	one variable missing

What we call it	Math term	Notation
hyperboloid (1 sheet)	hyperboloid of one sheet	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$
hyperboloid (2 sheets)	hyperboloid of two sheets	$-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$
cone	elliptic cone	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$